

1-1 Prime Factorization

Number System Launch | Standard: 6.NS.4 | Color-by-Number Practice

Learning Goal: I can write a number as a product of its prime factors using a factor tree.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: prime, factor, factor tree, product

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Write the prime factorization of 12. Work: _____ Answer: _____	2. Write the prime factorization of 18. Work: _____ Answer: _____
3. Write the prime factorization of 20. Work: _____ Answer: _____	4. Write the prime factorization of 28. Work: _____ Answer: _____
5. Write the prime factorization of 30. Work: _____ Answer: _____	6. Write the prime factorization of 36. Work: _____ Answer: _____
7. Write the prime factorization of 42. Work: _____ Answer: _____	8. Write the prime factorization of 45. Work: _____ Answer: _____
9. Write the prime factorization of 50. Work: _____ Answer: _____	10. Write the prime factorization of 63. Work: _____ Answer: _____
11. Write the prime factorization of 72. Work: _____ Answer: _____	12. Write the prime factorization of 84. Work: _____ Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

1-1 Mystery Picture: Mission Patch Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
$2 \times 2 \times 3 =$ red	$2 \times 3 \times 3 =$ orange	$2 \times 2 \times 5 =$ yellow	$2 \times 2 \times 7 =$ green
$2 \times 3 \times 5 =$ blue	$2 \times 2 \times 3 \times 3 =$ purple	$2 \times 3 \times 7 =$ pink	$3 \times 3 \times 5 =$ brown
$2 \times 5 \times 5 =$ gray	$3 \times 3 \times 7 =$ teal	$2 \times 2 \times 2 \times 3 \times 3 =$ black	$2 \times 2 \times 3 \times 7 =$ light blue

2	2	1	2	2	2	3	3	3	4	4	6
2	5	5	5	5	5	9	6	6	7	9	6
1	5	8	9	8	8	9	9	9	12	9	6
7	6	10	2	2	1	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	10	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	11	3	6	12	1	10	4	2	11	8
5	2	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	4	5	3	11	8
7	10	3	4	4	2	3	3	4	5	2	10
10	12	12	1	11	11	12	12	12	6	2	11
9	9	9	8	8	8	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	$2 \times 2 \times 3$	red	Use a factor tree until every branch ends in a prime number.
2	$2 \times 3 \times 3$	orange	Use a factor tree until every branch ends in a prime number.
3	$2 \times 2 \times 5$	yellow	Use a factor tree until every branch ends in a prime number.
4	$2 \times 2 \times 7$	green	Use a factor tree until every branch ends in a prime number.
5	$2 \times 3 \times 5$	blue	Use a factor tree until every branch ends in a prime number.
6	$2 \times 2 \times 3 \times 3$	purple	Use a factor tree until every branch ends in a prime number.
7	$2 \times 3 \times 7$	pink	Use a factor tree until every branch ends in a prime number.
8	$3 \times 3 \times 5$	brown	Use a factor tree until every branch ends in a prime number.
9	$2 \times 5 \times 5$	gray	Use a factor tree until every branch ends in a prime number.
10	$3 \times 3 \times 7$	teal	Use a factor tree until every branch ends in a prime number.
11	$2 \times 2 \times 2 \times 3 \times 3$	black	Use a factor tree until every branch ends in a prime number.
12	$2 \times 2 \times 3 \times 7$	light blue	Use a factor tree until every branch ends in a prime number.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.